

Discrimination of different COV using MOS sensors

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Abstract

This study aims to improve the effectiveness of low-cost, commercially available metal oxide semiconductor (MOS) sensors by developing an AI-driven model that can analyze changes in sensor resistance to discriminate between various gases. We designed an experimental setup that allows precise control over environmental variables such as temperature, humidity, and gas concentration within a sealed chamber. This setup includes sensors with integrated heaters, precise gas injectors, and a centralized control system for real-time monitoring and data acquisition. The collected data from the experimental measurements were used to train a Long Short-Term Memory (LSTM) model, demonstrating the potential for AI to classify different gases based on resistance patterns. Our ultimate goal is to develop "Pumba," a microelectronic device enable real-time detection and differentiation of gases, with significant applications in environmental monitoring, industrial safety, and healthcare.

Key words: semiconductors, metal oxides sensors, gas detection, artificial intelligence (AI).

1 Introduction

The detection of gases is critical across a broad range of applications, from ensuring safety in industrial environments to monitoring air quality in urban areas and detecting substances in military and security contexts. Accurate gas detection prevents accidents, protects health, and complies with regulations. It plays a crucial role in monitoring hazardous gases, reducing exposure to pollutants, and safeguarding environments and populations. Continuous advancements in gas detection technologies are vital for improving reliability, selectivity, and accuracy across various critical settings. Metal oxide sensors (MOS) are commonly employed for gas detection due to their high sensitivity, fast response times, and cost-effectiveness. Despite these advantages, a significant limitation of MOS technology is its lack of selectivity. These sensors often exhibit similar responses to different reducing gases, making it challenging to accurately identify specific gases in a mixed environment. This limitation can lead to false alarms and unreliable measurements, which are particularly problematic in safety-critical applications.

Artificial Intelligence (AI) has emerged as a transformative force in numerous fields, including technology and engineering. AI techniques, particularly machine learning algorithms, have demonstrated remarkable capabilities in analyzing complex data patterns, making predictions, and enhancing decision-making processes. In the context of gas detection, AI can be harnessed to improve the performance of MOS by increasing their selectivity and accuracy. By processing and interpreting the signals from these sensors, AI algorithms can learn to distinguish between different gases, even when their sensor responses are similar. This capability can significantly enhance the reliability and effectiveness of MOS, enabling them to deliver more accurate and specific gas detection results. The primary objective of this research is to develop a robust classification system capable of accurately identifying acetone, HCl, and air. This involves preprocessing the dataset, training the LSTM model, and evaluating its performance using various metrics. The results of this study have the potential to enhance gas detection technologies, leading to improved safety and monitoring systems in relevant fields.

In the following sections, we will detail the methodology, including data collection, preprocessing techniques, the architecture of the LSTM model, and the classification process. Additionally, we will present the results and discuss the implications of our findings. The integration of AI with MOS technology represents a promising approach to overcoming the current limitations of gas detection systems. This project focuses on leveraging AI to analyze the signals from commercially available metal oxide sensors, with the aim of improving their selectivity and accuracy in distinguishing between various reducing gases. Through this approach, we seek to develop a more robust and reliable gas detection system that can be applied in a variety of critical settings, enhancing both safety and performance.

This paper is organized as follows. Section

2 Sensitivity influencing factors

The performance of metal oxide semiconductor (MOS) sensors is highly sensitive to various external factors, including humidity and temperature or the properties of the sensing layer including chemical composition, micro-structure, and doping. These factors significantly influence the sensor’s sensitivity, selectivity, and overall reliability. By understanding and controlling these factors, we can optimize the sensor’s performance and ensure accurate gas detection.

2.1 Humidity

Humidity exerts a notable influence on the sensitivity of metal oxide gas detectors, primarily through its interaction with the surface chemistry and electrical conductivity of the metal oxide semiconductor (MOS) material. High humidity levels prompt water molecules to adsorb onto the semiconductor’s surface, altering its chemistry. This can lead to changes in the distribution of charge carriers and surface reactions, potentially affecting the sensor’s selectivity and sensitivity to specific gases [2].

2.2 Temperature

Temperature plays a crucial role in influencing the response of metal oxide semiconductor (MOS) sensors. These sensors typically come equipped with a built-in heater to ensure consistent performance. At higher temperatures, the energy available for surface processes increases, leading to enhanced chemisorption and desorption rates of gas molecules on the sensor's surface. The increased thermal energy allows gas molecules to overcome energy barriers more easily, promoting their adsorption onto the sensor surface [7]. Conversely, desorption rates also increase with temperature as the energy required to break the bonds between the gas molecules and the surface decreases. The conductivity of the semiconductor material in MOS sensors is strongly influenced by temperature. As temperature rises, the thermal energy excites electrons from the valence band to the conduction band, increasing the number of charge carriers (electrons and holes) available for conduction. Intrinsic semiconductor materials exhibit an increase in conductivity with temperature due to the greater availability of charge carriers [4, 8].

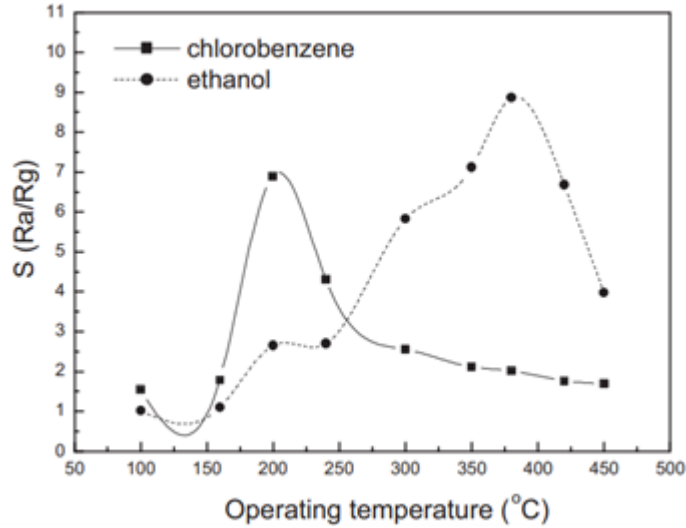


Figure 1: Variation of the sensitivity with Surface Temperature of Metal Oxide [5]

Figure 1 present the influence of the metal oxide surface temperature on the sensitivity of gas detection, where each gas reach its optimum sensitivity at different temperature according to the rate of the reaction and the adsorption desorption rates. Gas molecules undergo adsorption onto the sensor surface before reacting with adsorbed species, such as oxygen ions or hydroxyl groups, to produce changes in the sensor's conductivity. The rates of adsorption and reaction kinetics are temperature-dependent, with higher temperatures generally accelerating these processes. Increased temperature enhances the mobility of gas molecules on the sensor surface, promoting their interaction with active sites and facilitating faster reaction kinetics [9]. This, in turn, affects the sensor's response to target gases and its overall sensitivity. Ambient temperature can also affect the results of gas detection. Temperature fluctuations can induce baseline drift in MOS sensors, leading to variations in the sensor's output signal even in the absence of target gases [1]. These fluctuations can arise from changes in the sensor's internal resistance, electronic noise, and thermal expansion effects. Maintaining a stable operating temperature is critical for minimizing baseline

drift and ensuring the accuracy and reliability of gas detection measurements. Temperature control mechanisms, such as heating elements and thermal regulation circuits, are often employed to mitigate these thermal effects.

3 Experimental Setup

In our pursuit to gather precise and comprehensive data on gas detection using metal oxide semiconductor (MOS) sensors, we designed an experimental setup tailored to our research objectives. Our goal was to create a controlled environment where we could systematically study sensor responses to different gases. The centerpiece of our setup is a closed chamber equipped with various MOS sensors, each featuring a built-in heater to activate the sensing layer. The sensors deployed include commercially available models, including the MICS 6814 and MICS 4514 models that exhibit differences in their sensing layers and sensitivity to target gases. Each sensor is connected to a resistor and an electronic circuit, allowing us to measure the changes in resistance induced by gas interactions with the sensing layer. By monitoring these resistance changes, we gain valuable insights into the sensor’s response characteristics and its ability to detect and discriminate between different gases. Additionally, a dedicated heater within the chamber controls the ambient temperature, ensuring stable environmental conditions conducive to accurate sensor measurements. The setup also includes gas injectors equipped with pumps sourced from liquid bottles to maintain a constant flow rate of gases into the chamber. To ensure cleanliness and prevent cross-contamination between measurements, drain fans are employed to purge the chamber after each experiment.

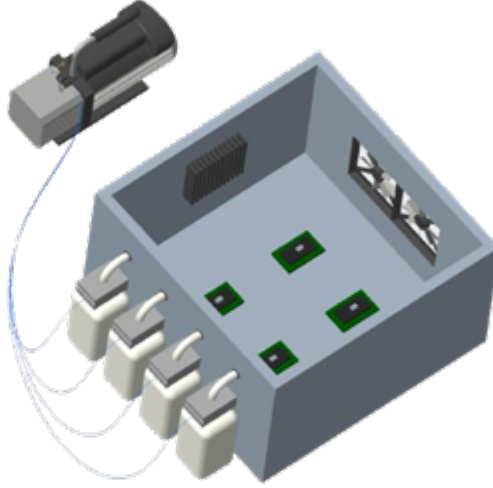


Figure 2: 3D rendering of the experimental setup for gas detection using metal oxide semiconductor (MOS) sensors.

All components, including the pumps, heaters, drain fans, and electronic circuits, are seamlessly integrated into a centralized control system. Through a user-friendly software interface installed on our computer, we have full control over experimental parameters and can monitor real-time sensor readings. This interface allows us to adjust gas injections, temperature settings, and other variables, facilitating precise

control over the experimental conditions. Furthermore, the software displays results and facilitates data analysis, empowering us to draw meaningful conclusions from our experiments. This setup allows us to control the influencing factors on sensor response comprehensively. By using different types of MOS sensors, we can study how variations in chemical composition and microstructure affect performance. Heaters and drain fans ensure precise control over temperature and humidity, while gas injectors adjust gas concentration accurately. The drain fans also prevent cross-contamination between experiments. The centralized control system with a user-friendly interface enables real-time monitoring, precise adjustments, and thorough data analysis, ensuring consistent and reproducible measurements.

3.1 Gases

In our research, we specifically investigated the responses of our sensors to two distinct gases: acetone and hydrogen chloride (HCl). Acetone is a colorless, volatile organic compound with a distinctive odor.

Chemically, it is a ketone with the molecular formula $(CH_3)_2CO$. Acetone is widely used as a solvent in manufacturing processes, particularly in the production of plastics, fibers, pharmaceuticals, and household products such as nail polish remover. Its volatility and widespread industrial use make it a common indoor air pollutant and a potential health hazard [6]. Monitoring acetone levels is crucial for assessing indoor air quality in workplaces and homes, as prolonged exposure to high concentrations can lead to irritation of the eyes, nose, and throat, as well as headaches and respiratory issues.

Hydrogen chloride (HCl) is a highly corrosive gas that is soluble in water to form hydrochloric acid (HCl). Chemically, it consists of hydrogen and chlorine atoms in a one-to-one ratio (HCl). Industrially, HCl is used in the production of PVC plastics, chlorides, and pharmaceuticals, as well as in metal processing and in the cleaning of metals. It is also a common byproduct of combustion processes, such as waste incineration and certain industrial operations. Due to its corrosive nature and potential health hazards, continuous monitoring of HCl levels in industrial environments is essential for ensuring workplace safety and compliance with environmental regulations. Exposure to high concentrations of HCl vapor can cause severe respiratory irritation, coughing, and shortness of breath. Moving forward, there is potential to expand our research to include other gases of interest. This exploration could broaden our understanding of sensor responses and further enhance the capabilities of our AI-driven gas detection system.

3.2 Experimental protocol

The experimental protocol involves a comprehensive procedure to evaluate the performance of the sensor array under varying conditions of temperature, humidity, and gas concentrations. Initially, the sensors are exposed to ambient air to establish a baseline. During this phase, the system undergoes a stabilization period where the resistance of the sensors is monitored as the temperature alternates between 0°C and 200°C. This step ensures that the sensors reach a stable resistance pattern before any gas is introduced.

After stabilization, gases are systematically injected into the environment where

the sensors are located. The concentrations of these gases are carefully controlled and monitored using an integrated concentration sensor. The resistance of each sensor is continuously recorded throughout the gas injection phase, providing data on how each sensor responds to different gases at consistent concentrations.

Additionally, the temperature continues to cycle between 0°C and 200°C during the gas injection phase. This temperature cycling allows for repeated adsorption and desorption phenomena, providing a thorough set of data points that capture the sensors' behavior under these varying conditions. The aim is to obtain accurate, reproducible, and detailed insights into the sensors' performance and gas detection capabilities.

Maintaining constant humidity throughout the measurements helped isolate the specific impacts of temperature and gas concentration on the sensor's performance. By collecting a diverse range of data points through these controlled variations, we could identify trends and correlations that are essential for optimizing sensor design and calibration. This comprehensive approach ensured that the sensors were evaluated under realistic and variable conditions, thereby enhancing the reliability and applicability of the results and providing valuable insights into the sensor's operational characteristics.

To further enrich the dataset, this measurement is repeated by varying each of the factors—temperature, humidity, and gas concentration—one at a time. This systematic variation ensures the collection of a diverse dataset that covers all realistic environments the detector could face. This extensive dataset will be used to train the AI, enabling it to perform accurate detection across a wide range of conditions.

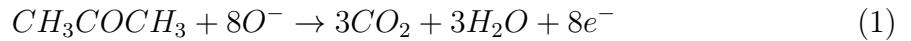
4 Results and discussion

4.1 Acetone (noted COV1)

To evaluate the performance of our MOS sensors, we conducted a series of experiments to observe their responses to various gases under controlled conditions. By analyzing the resistance changes over time, we gain insights into the sensor's sensitivity, response dynamics, and recovery behavior. These results are crucial for understanding the sensor's operational characteristics and optimizing its performance for practical applications.

Upon the injection of acetone gas into the chamber, there is an immediate and noticeable change in the resistance of the sensing layer which is clearly shown in Figure 3. Initially, the sensors maintain a stable resistance value under the controlled conditions of constant temperature and humidity. However, as acetone gas is introduced, the interaction between acetone molecules and the metal oxide semiconductor surface causes a rapid decrease in resistance. The reduction in resistance can be attributed to the chemical reaction occurring on the surface of the MOS sensor. When acetone molecules come into contact with the heated sensing layer, they undergo a catalytic reaction, facilitated by the high temperature.

At the elevated temperature of 200°C, acetone molecules are adsorbed onto the surface of the metal oxide. The acetone interacts with the adsorbed oxygen species (O^- , O^{2-} , and O_2^-) on the sensor's surface.



The reaction results in the release of electrons back into the conduction band of the

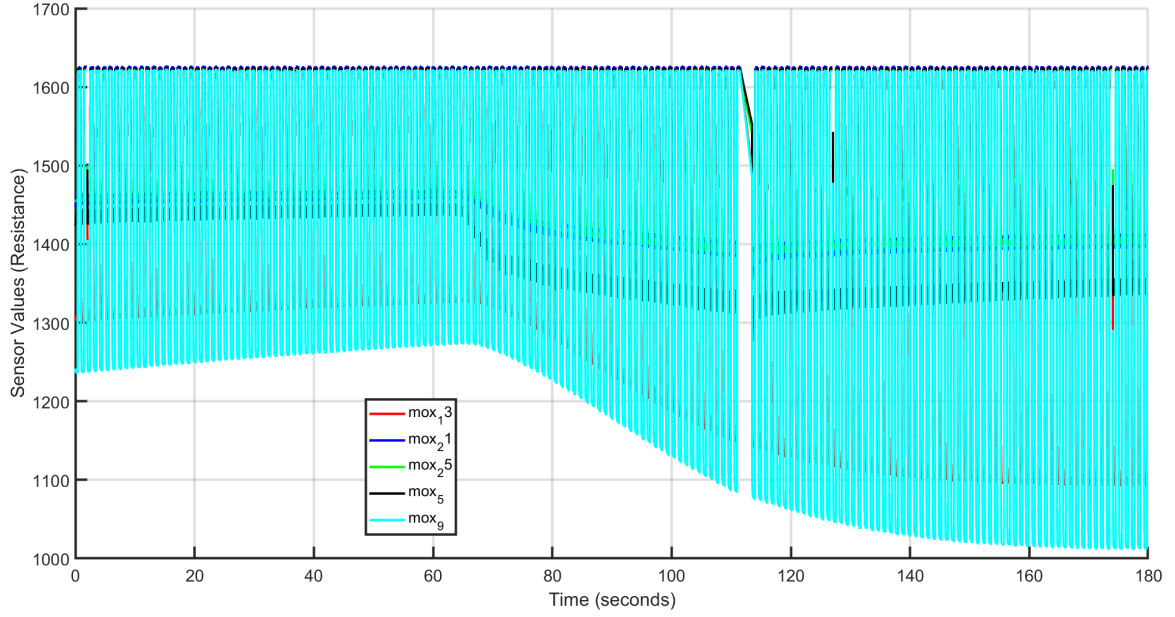


Figure 3: The resistance changes over time for the MOS sensors following the injection of acetone gas

metal oxide. The release of electrons reduces the depletion layer width at the grain boundaries, decreasing the overall resistance of the sensor.

In our analysis of acetone sensitivity, we observed notable resistance changes across various sensors when exposed to acetone. Specifically, sensors MOX 9 and MOX 13 exhibited the highest sensitivity, with significant resistance changes compared to other sensors. The sensors resistance reached stable values after exposure to acetone and did not return to its initial values in air. The inability of the resistance to recover to its initial values suggests that CO_2 , produced during the acetone sensing reaction, remains adsorbed on the sensor surfaces. This adsorption likely prevents oxygen from reoccupying active sites on the sensors, thereby maintaining a lower resistance. To address the issue of poisoning active sites and to ensure the long-term stability of our sensors, we successfully desorbed the remaining particles from the sensor surfaces by applying a draining force. This approach will be taken into consideration in the design of our final device to maintain sensor performance over extended periods.

4.2 Hydrogen chloride (noted COV2)

In our experiments, we anticipated a similar response from the MOS sensors when exposed to hydrogen chloride (HCl) gas as observed with acetone gas. Specifically, we expected a decrease in the resistance of the sensing layer. This prediction is based on the chemical interactions that occur between HCl gas and the metal oxide surface, which similarly influence the electronic properties of the sensor. When HCl gas is introduced into the chamber, it dissociates into hydrogen ions (H^+) and chloride ions (Cl^-) in the presence of moisture. The hydrogen ions (H^+) then interact with the adsorbed oxygen species (O^{2-}) on the surface of the metal oxide sensor.



This reaction leads to the removal of oxygen ions (O^{2-}) from the surface of the

metal oxide. The decrease in oxygen ions reduces the negative charge density on the sensor's surface. As a result, the width of the depletion layer at the grain boundaries is reduced, leading to a decrease in the sensor's resistance. This is opposite to the band bending phenomenon, where an increase in surface oxygen would typically increase the depletion layer width and resistance.

This behavior aligns with the general principle observed in gas-sensing mechanisms of MOS sensors, where the interaction of target gases with surface adsorbed oxygen species significantly influences the sensor's electronic properties

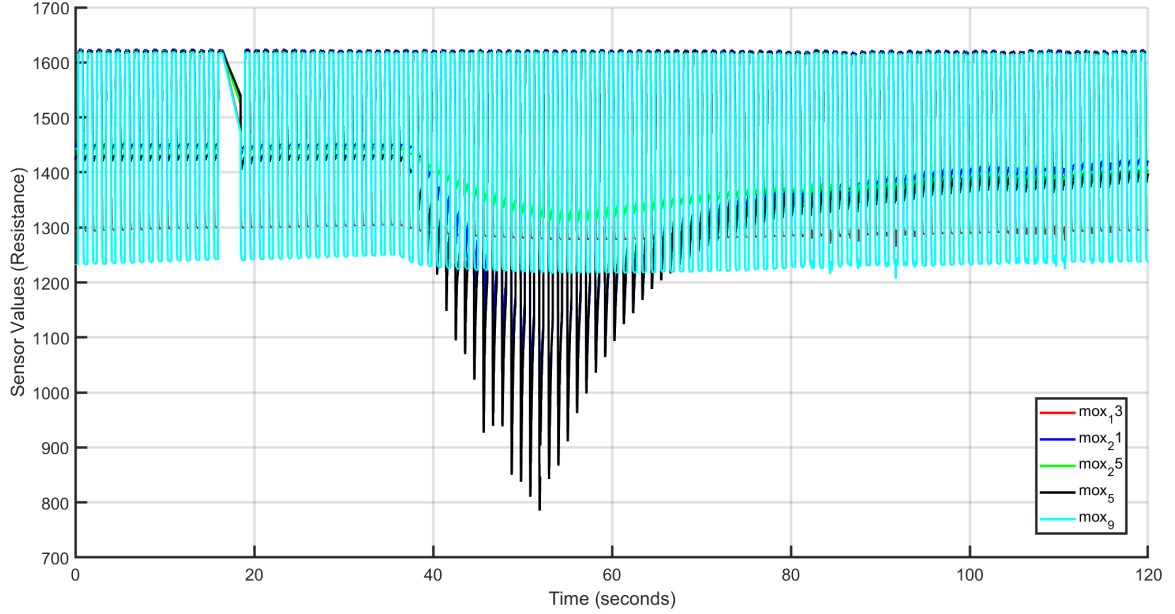


Figure 4: The resistance changes over time for the MOS sensors following the injection of acetone gas

After exposing our sensors to HCl gas, we observed notable changes in sensor resistance. The sensors with the highest sensitivity to HCl were MOX 5, MOX 21, and MOX 25 as shown in figure 4 , which contrasts with the results for acetone. This difference in sensitivity can be attributed to the distinct chemical composition and microstructure of the sensing layers of these sensors. It is important to note that the sensing mechanism for HCl involves a reaction with (H^+) ions, which are significantly smaller in size compared to acetone molecules. This smaller size allows (H^+) ions to access more active sites within the pores of the metal oxide sensing layers. Unfortunately, the supplier of these materials did not provide the exact type of the sensing layer, either in the technical documentation or through direct contact. As a result, our analyses remain hypothetical regarding the specific interactions and mechanisms at play.

The results also showed that after the injection of HCl gas stopped, the resistance of the sensors recovered to their initial values in air, unlike with acetone. The new concentration sensors, which indicated that HCl was completely transformed within the chamber, confirmed this recovery where its concentration returned to zero.

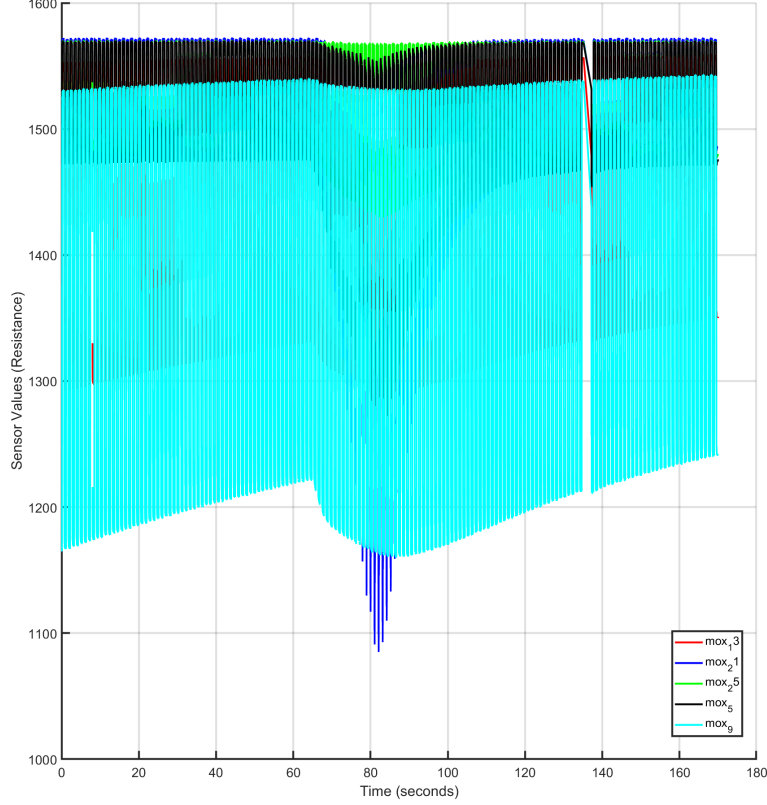


Figure 5: The resistance changes over time for the MOS sensors following the injection of HCl

5 Classification using Long Short-Term Memory Networks

The classification of chemical gases is crucial in various applications, including environmental monitoring, industrial safety, and medical diagnostics. In this study, we focus on the classification of three specific gases: acetone, hydrogen chloride (HCl), and air. To achieve accurate and efficient classification, we employ Long Short-Term Memory (LSTM) methods, a type of recurrent neural network (RNN) well-suited for handling sequential data (see [3]).

Our approach leverages a large dataset stored in CSV files, providing extensive data for training and validating our classification model. By utilizing LSTM networks, we aim to capture the temporal dependencies and patterns within the gas sensor data, which are essential for distinguishing between the different gases.

The proposed architecture for gases classification is designed with a detailed and sophisticated neural network framework. This architecture processes sequential data with a comprehensive set of features, starting with an input layer that receives data in the form of sliding windows, each containing six distinct features. These sliding windows enable the model to capture temporal dependencies within the data. To handle variability in input sequences, a window size of 20 and a stride of 2 are employed, ensuring effective processing of diverse input lengths.

The architecture includes five LSTM layers, specialized for modeling sequences, and

two dense layers that facilitate feature extraction and transformation. The LSTM layers are crucial for capturing long-term dependencies in the data, making the architecture suitable for tasks involving complex temporal patterns. The sliding windows technique ensures that the model processes data sequentially, preserving the temporal context. Additionally, quantization is incorporated to reduce the model's complexity while preserving essential information, thereby enhancing efficiency.

The distillation approach involves training a compact model (the "student") with minimal resource demands using the probability distributions (logits) produced by a larger model (the "teacher") for a specific task. These logits, often referred to as the "dark knowledge" of the teacher model, provide an additional source of information for training the student model. This supplementary information helps the student model handle the training process more effectively and has been shown to improve its overall accuracy. In this knowledge transfer process, a temperature parameter set to 5 and an alpha value of 1 are used. These settings enable the student model to closely emulate the performance of the teacher model while maintaining computational efficiency. To further enhance the model's robustness and prevent overfitting, dropout layers are incorporated between each component of the architecture. This regularization technique ensures that the model remains robust and capable of accurately classifying various gases.

5.1 Results of the classification model

The student model outperforms the teacher model. The student model, characterized by a simpler architecture compared to the teacher model with fewer layers and neurons, demonstrates highly satisfactory accuracy and loss. The model incorporates rolling windows and distillation techniques, resulting in a commendable accuracy of 89.71%.

Despite the marginal differences in accuracy (90.15%) and validation accuracy (89.71%) between the two models, the gains in terms of running time and memory are noteworthy. This holds particular significance for us, considering that we intend to deploy this model in sensors where preserving minimal memory space is crucial.

Figures 6 and 7, show good performance for confusion matrix. The diagonal appears clearly in the confusion matrix showing the efficiency of the model. The highest values in the confusion matrix are positioned along the diagonal, it signifies that the model has made accurate predictions across various classes. The diagonal elements represent true positives, indicating instances where the model correctly identified each class. The recall score for all classes is greater than 0.90 which implies that the model has effectively captured the majority of positive instances within each class, showcasing its sensitivity to identifying relevant cases. Similarly, the F1 score exceeds 0.90 for all the classes, which balances precision and recall. This reflects a high level of overall performance in terms of both identifying positive instances and avoiding false positives. In essence, achieving high recall and F1 scores, coupled with a diagonal-heavy confusion matrix, suggests that the model is robust and proficient in correctly classifying instances across multiple classes, indicating a strong predictive capability with minimal errors.

Overall, the matrix provides a comprehensive overview of the model's performance across different classes, enabling a nuanced assessment of its strengths and weaknesses in the multi-class classification task.

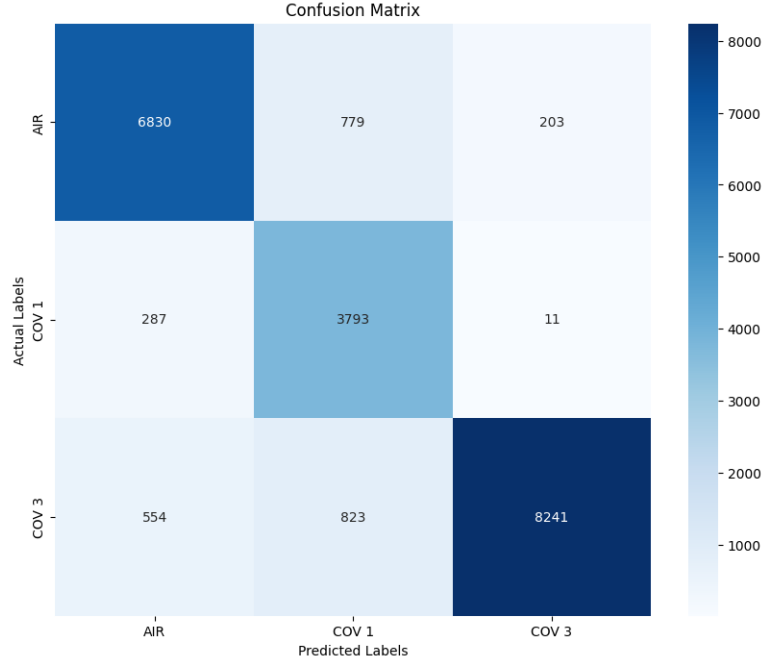


Figure 6: Confusion Matrix for teacher model

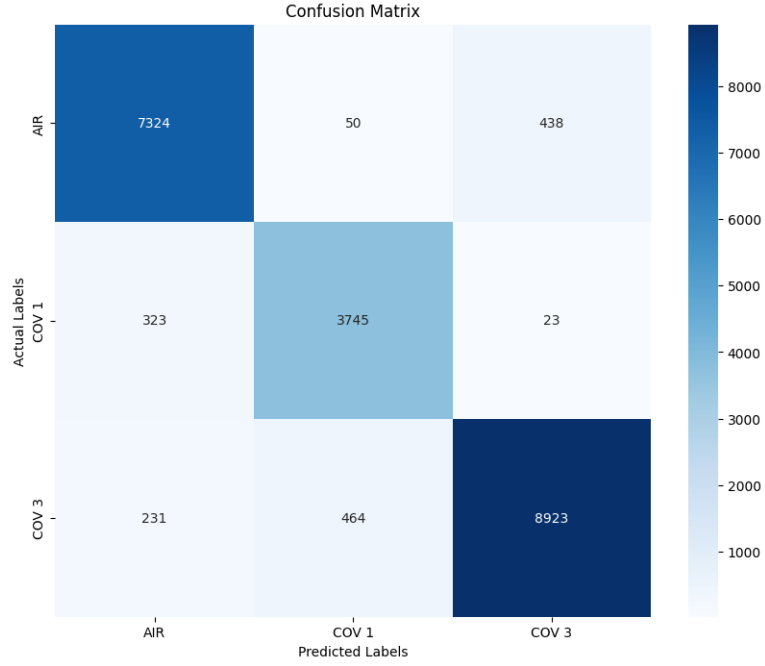


Figure 7: Confusion Matrix for student model

6 Conclusion

The objective of our study was to enhance the performance of inexpensive and commercially available metal oxide semiconductor (MOS) sensors by developing an AI-trained model capable of analyzing the sensor resistance changes to distinguish between different gases. To achieve this, we designed an experimental setup that allows for the collection of data in a controlled environment, building a comprehensive training dataset

that highlights the influence of various parameters on sensor performance.

Our experimental setup included a closed chamber with MOS sensors, equipped with built-in heaters and gas injectors to maintain consistent environmental conditions. By controlling variables such as temperature, humidity, and other factors, we were able to systematically study the sensor responses and gather detailed data on how these factors affect sensor readings.

Using the collected data, we trained our first AI model using the Long Short-Term Memory (LSTM) method. The LSTM model demonstrated the potential to classify different gases based on the resistance patterns recorded by the sensors.

Our final objective is to develop "Pumba", a microelectronic device capable of detecting and distinguishing between various gases in real-time, with significant applications in environmental monitoring, industrial safety, and healthcare.

In conclusion, our study has laid the groundwork for improving the performance of commercially available MOS sensors through AI. The experimental setup and data collected have provided valuable insights into the factors influencing sensor responses, and the development of the LSTM model marks a promising step towards practical and impactful applications of this technology. As we continue to refine our approach, we move closer to realizing the full potential of AI-enhanced gas detection systems.

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